

TEST REPORT

PROJECT: PetCare+

Comprehensive Test Report

Version: 1.1

Date: November 18, 2025

Status: PASSED (Conditional)

1. Executive Summary

This report provides a comprehensive overview of all testing activities performed on the PetCarePlus application. The testing strategy encompasses multiple layers including unit tests, integration tests, API tests, and end-to-end (E2E) tests using both Selenium and Playwright frameworks.

1.1 Test Coverage Summary

Test Type	Total Tests	Passed	Failed	Skipped	Coverage
Unit Tests	156	156	0	0	87.3%
Integration Tests	89	89	0	0	76.5%
API Tests	67	67	0	0	92.1%
E2E Tests (Selenium)	34	33	1	0	85.7%
E2E Tests (Playwright)	28	28	0	0	100%
TOTAL	374	373	1	0	88.3%

1.2 Overall Test Metrics

- Total Test Execution Time: 12 minutes 47 seconds
- Build Status: PASSING
- Code Coverage: 88.3%
- Critical Path Coverage: 95.2%
- Performance Tests: All within SLA

2. Unit Tests

2.1 Test Structure

Unit tests are organized by layer following Clean Architecture principles.

Directory Structure:

- backend/tests/PetCare.Domain.Tests/ (32 tests)
- backend/tests/PetCare.Application.Tests/ (68 tests)
- backend/tests/PetCare.Infrastructure.Tests/ (36 tests)
- backend/tests/PetCare.Api.Tests/ (20 tests)

2.2 Domain Layer Tests

Status: 100% Pass Rate (32 Tests)

Test Suite	Tests	Result	Coverage
UserTests	8	Passed	100%
PetTests	6	Passed	100%
AppointmentTests	9	Passed	100%

Sample Scenarios:

- User creation with default values.
- Pet age calculation based on date of birth.
- Appointment status transitions.

2.3 Application Layer Tests

Status: 100% Pass Rate (68 Tests)

Critical Path Tests Included:

- User registration with password hashing
 - JWT token generation and validation
 - Appointment approval workflow with notifications
-

3. Integration Tests

Total Tests: 89

Success Rate: 100%

Average Execution Time: 2.3s

3.1 API Integration Scenarios

Scenario A: Complete Appointment Workflow

- 1. Owner registers account → 201 Created
- 2. Owner logs in → 200 OK (JWT received)
- 3. Owner requests appointment → 201 Created (Pending)
- 4. Admin approves appointment → 200 OK
- 5. Notification verified in database

Result: PASSED

Scenario B: Medical Record Creation

- 1. Vet creates medical record → 201 Created
- 2. Vet adds prescription → 201 Created
- 3. Owner retrieves history → 200 OK

Result: PASSED

4. End-to-End Tests (Selenium)

Framework: Selenium WebDriver (C#)

Browsers: Chrome, Firefox, Edge

4.1 Workflow Results

Test Suite	Tests	Passed	Failed	Duration
Owner Workflows	12	12	0	3m 24s
Admin Workflows	10	10	0	2m 56s
Vet Workflows	7	7	0	2m 12s
Cross-Role Scenarios	5	4	1	1m 48s

4.2 Failed Scenario Analysis

Test: MultipleSimultaneousBookings

Description: Two owners book the same time slot simultaneously.

Expected: Second booking should fail.

Actual: Both bookings succeeded.

Issue ID: DEF-001 (JIRA-1247)

Priority: HIGH

5. End-to-End Tests (Playwright)

Framework: Playwright (TypeScript)

Total Tests: 28

Status: 100% Passed

5.1 Infrastructure & Advantages

- Browsers: Chromium, Firefox, WebKit
- Speed: 48% faster than Selenium
- Features: Auto-waiting, network interception, video on failure

5.2 Key Playwright Scenarios

Visual Regression

- Homepage layout
 - Dashboard rendering
- Result:** All matched baseline.

Responsive Design

Tested on desktop, tablet, and mobile.

Result: All layouts passed.

Network Condition Testing

Simulated fast 3G, slow 3G, offline.

Result: No instability encountered.

Accessibility Testing

Verified ARIA labels, keyboard navigation.

Result: WCAG 2.1 AA compliant.

5.3 Browser Coverage Results

Browser	Tests	Passed	Failed	Avg Duration
Chromium	28	28	0	10.8s
Firefox	28	28	0	11.4s
WebKit	28	28	0	12.1s

6. Performance Testing

Tool: JMeter

Environment: Staging (4 CPU, 8GB RAM)

6.1 Load Test Summary

Scenario	Users	Duration	Success Rate	Avg Response
Normal Load	50	10 min	99.8%	142ms
Peak Load	200	5 min	98.2%	387ms
Stress Test	500	3 min	92.4%	1,234ms

Findings:

- System stable under normal load
- Connection pool exhausted at 450+ users
- Recommendation: Increase DB pool size + introduce caching